

Till now we understood theory...
Today we build our first Neural Network.”

👉 Pause

Then say:

“This is the same foundation used in real AI systems.”

2. 📖 WHAT IS TENSORFLOW?

Definition

TensorFlow is a Deep Learning framework developed by Google.

Use

Used for:

- Neural Networks
 - Deep Learning
 - AI applications
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3. 📖 WHAT IS KERAS?

Definition

Keras is a high-level API for building deep learning models easily.

Simple Meaning

“Keras makes Deep Learning easier.”

4. 📖 WHY USE KERAS?

- ✓ Simple syntax
 - ✓ Fast model building
 - ✓ Beginner friendly
 - ✓ Built on TensorFlow
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5. 📖 ANN BUILDING FLOW

Complete Pipeline

1. Import libraries
 2. Create dataset
 3. Define X and y
 4. Create ANN model
 5. Compile model
 6. Train model
 7. Evaluate model
 8. Predict new output
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6. IMPORTANT FUNCTIONS

Function	Purpose
Sequential()	Create ANN
Dense()	Create layers
compile()	Configure learning
fit()	Train model
evaluate()	Test model
predict()	Make predictions

7. DENSE LAYER

Meaning

Fully connected neural network layer.

Example

```
Dense(4, activation='relu')
```

Means:

- 4 neurons
 - ReLU activation
-
-

8. COMPILE STEP

Purpose

Defines:

- optimizer
 - loss function
 - evaluation metric
-

Example:

```
optimizer='adam'  
loss='binary_crossentropy'  
metrics=['accuracy']
```

9. FIT (TRAINING)

Meaning

Model learns from data.

Important Parameters

Parameter Meaning

epochs learning cycles

batch_size rows processed together

10. EVALUATION

Checks:

- loss
 - accuracy
-
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11. PREDICTION

Model predicts new unseen data.

12. PRACTICAL (YOUR FORMAT)

Objective

Build first ANN classification model

OUTPUT INTERPRETATION

Training Output

Students will see:

- epochs
 - loss
 - accuracy
-

Prediction Output

Example:

[[0.82]]

Meaning:

- 82% probability of passing
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If threshold = 0.5:

👉 Final class = Pass

14. EXPLAIN MODEL STRUCTURE

Hidden Layer

Dense(4, activation='relu')

Means:

- 4 neurons
 - learns patterns
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Output Layer

Dense(1, activation='sigmoid')

Means:

- binary classification output
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15. 📖 WHY RELU + SIGMOID?

Layer Activation

Hidden ReLU

Output Sigmoid

Reason

- ReLU learns efficiently
 - Sigmoid gives probability
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16. 📖 WHY ADAM OPTIMIZER?

Because:

- ✓ fast
 - ✓ stable
 - ✓ widely used
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17. ⚠️ IMPORTANT TEACHING NOTE

Tell students:

“Deep Learning usually needs large datasets.
Here we are learning workflow using small data.”

18. 🎯 INTERVIEW CONCEPTS

Q1: What is TensorFlow?

👉 Deep Learning framework.

Q2: What is Keras?

👉 High-level API for neural networks.

Q3: What does Dense layer mean?

👉 Fully connected layer.

Q4: Why sigmoid in output layer?

👉 Gives probability between 0 and 1.

Q5: What does epoch mean?

👉 One full pass through dataset.

“Artificial Neural Networks learn patterns by training interconnected neurons through repeated learning cycles.”